

Superoxide dismutase activity assessment

Material:

1. Sample
2. Potassium phosphate buffer
3. 2 mM pyrogallol

Procedures:

1. Homogenize the sample in a sonicator (Branson Sonifier Model 250; three 4-s cycles) in potassium phosphate buffer 100 mM and EDTA 50 mM, pH 7.4, and centrifuge the homogenate at 10.000 rpm and 4 °C for 10 min.
2. Incubate the spinal cord segment homogenate in a solution containing potassium phosphate buffer 100 mM and EDTA 50 mM, pH 7.4.
3. Reaction is initiated by the addition of 2 mM pyrogallol.
4. Measure the pyrogallol oxidation at 420 nm (UV/visible U-200L Spectrophotometer, Hitachinaka, Japan) for 5 min, at 30-s intervals. A 50 % inhibition was defined as one unit (U) of SOD.
5. The results are normalized by protein content and expressed as a percentage of the placebo group.